EQUIVALENCE OF REGULARITY, NORMALITY, DISCRETE VALUATION RING, INVERTIBLE IDEAL SHEAF AND CARTIER DIVISOR CONDITIONS AT A CLOSED POINT OF A CURVE

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Allowed standard prerequisites (editorial).

- Effective Cartier divisors in points and the characterization of discrete valuation rings. Stacks Project, Divisors, Lemma divisors-lemma-effective-Cartier-in-points (tag 0B3C); Algebra, Lemma algebra-lemma-characterize-dvr (tag 034Z). The first identifies the divisor conditions with principality of the maximal ideal and the second gives the equivalence with regularity and normality.
- Density of the smooth locus and separability criteria for smoothness. Stacks Project, Varieties, Lemma lemma-dense-smooth-open-variety-over-perfect-field (tag 0B2J); Algebra, Lemma algebra-lemma-separable-smooth (tag 030V). Supply the smoothness equivalence under the perfect and separable hypotheses.

Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AG8: sources/stacks-chow/tags.txt, source lines 13186–13186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B8X: sources/stacks-chow/tags.txt, source lines 14155–14155. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Lemma 0.1. *Let k be a field. Let X be a curve over k. Let $x \in X$ be a closed point. We think of x as a (reduced) closed subscheme of X with sheaf of ideals $\mathcal{I}$. The following are equivalent*

(1) *$\mathcal{O}_{X,x}$ is regular,*
(2) *$\mathcal{O}_{X,x}$ is normal,*
(3) *$\mathcal{O}_{X,x}$ is a discrete valuation ring,*
(4) *$\mathcal{I}$ is an invertible $\mathcal{O}_X$-module,*
(5) *x is an effective Cartier divisor on X.*

If k is perfect or if $\kappa(x)$ is separable over k, these are also equivalent to

(6) *$X \to \mathrm{Spec}(k)$ is smooth at x.*

Proof. Since X is a curve, the local ring $\mathcal{O}_{X,x}$ is a Noetherian local domain of dimension 1 (Lemma 0A21). Parts (4) and (5) are equivalent by definition and are equivalent to $\mathcal{I}_x = \mathfrak{m}_x \subset \mathcal{O}_{X,x}$ having one generator (Divisors, Lemma 0AG8). The equivalence of (1), (2), (3), (4), and (5) therefore follows from Algebra, Lemma 00PD. The final statement follows from Lemma 0B8X in case k is perfect. If $\kappa(x)/k$ is separable, then the equivalence follows from Algebra, Lemma 00TV. □